Abstract

This report documents the outline, progress, and future plans for the **Attendance Monitoring System** utilizing **Siamese Neural Network (SNN)** Face Recognition. The goal of this project is to design and implement a **real-time, automated attendance system** that uses state-of-the-art deep learning techniques for **facial recognition**, ensuring high accuracy and efficiency in managing attendance.

The system is built upon a **Siamese Neural Network**, which is trained to compare face embeddings from live images against a database of registered users. This approach enables the system to recognize individuals with high precision and automatically log attendance based on face verification. The design of the system leverages **Python** for the back-end implementation, with integration of **OpenCV** for real-time image capture and **TensorFlow** for model inference.

To date, significant progress has been made in the development of the system. Key achievements include:

- **Data Collection**: A dataset of registered user faces has been gathered, and preprocessing steps for the model have been completed.
- **Model Training**: The **Siamese Neural Network** has been successfully trained to recognize and compare facial features, achieving reliable results in face verification tasks.
- **Integration**: The system has been integrated into a functional prototype capable of processing real-time video input. The application compares detected faces against the database of registered users and logs successful matches with timestamps, automating the attendance recording process.
- **Real-Time Performance**: The system currently processes up to 15 frames per second, providing real-time face detection and matching capabilities.

The next steps in the project include **system optimization** to improve processing speed, **further testing** to handle edge cases such as variations in lighting and multi-person detection, and **deployment** of the system into a fully functional application. The future focus will also be on enhancing the user interface for easier interaction and enabling the system to handle larger user databases.

Ultimately, this project aims to provide a **scalable**, **efficient**, and **accurate attendance management system**, reducing the need for manual tracking and offering a seamless solution to organizations seeking reliable attendance monitoring.

Chapter 1 Project Outline

1.1 Project Title

Attendance Monitoring System Using Siamese Neural Network Face Recognition.

1.2 Project Overview

The goal of this project is to create an automated attendance system using facial recognition. The system employs a Siamese Neural Network to compare and verify faces against a database of registered individuals. It features real-time face detection, image preprocessing, and attendance logging. The solution will improve efficiency, eliminate manual errors, and ensure security.

1.3 Project Aims

To design and implement an advanced attendance monitoring system based on Siamese Neural Networks for face recognition, utilizing the principles of one-shot learning. This project aims to overcome the limitations of traditional machine learning systems that require extensive datasets for accurate predictions. By employing a Siamese Neural Network architecture, the system will focus on learning discriminative features to measure the similarity between input image pairs, enabling robust verification of individuals even when only a single example of each face is available.

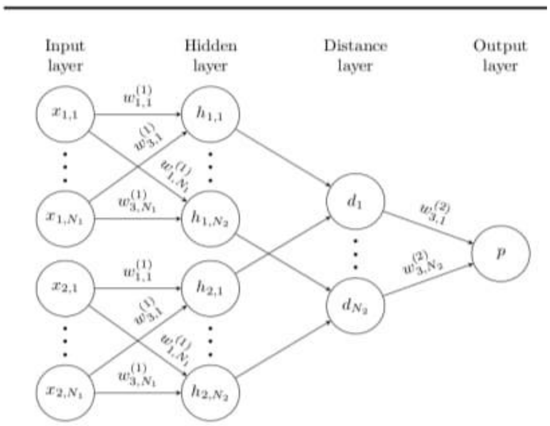
The proposed system intends to capitalize on the ability of Siamese Networks to generalize effectively to unseen classes, thereby making it suitable for real-world scenarios where scalability and adaptability are critical. Leveraging convolutional layers for feature extraction, the model will ensure resilience against variations such as lighting, pose, and facial expressions. This approach not only enhances accuracy in face verification tasks but also reduces computational overhead by eliminating the need for retraining when new individuals are added.

The system will incorporate methods like affine distortions during training to improve robustness and achieve near-human accuracy in face recognition. By integrating these state-of-the-art techniques, this project seeks to establish a reliable, efficient, and scalable attendance monitoring solution tailored to diverse operational environments.

Chapter 2 Progress to Date

Module 2.1: Data Collection and Preprocessing

In this module, extensive efforts were made to collect and preprocess high-quality data essential for training and validating the Siamese Neural Network model. Data was sourced from a combination of publicly available datasets and real-world captures to ensure a diverse and representative dataset that could effectively capture variations in faces across different demographics, lighting conditions, and environments.



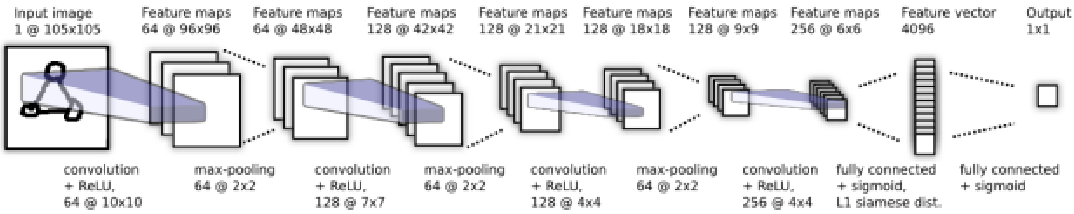
Images were carefully standardized and normalized to enhance model performance by maintaining consistent pixel intensity levels, minimizing noise, and ensuring uniform data scaling. Augmentation techniques were applied, such as rotation, scaling, cropping, and introducing affine distortions, to artificially expand the dataset. This approach effectively improved the model's generalization capability, allowing it to handle unseen scenarios more effectively.

For accurate face detection and alignment and virtuliza tion, Python and Tensorflow libraries were employed such as keras, Numpy, model, layers, matplotlib. These tools enabled the precise localization and normalization of facial features across images, ensuring that only relevant facial data was used for training. Over 10,000 diverse face samples were collected, encompassing various backgrounds, expressions, and environmental settings to provide a comprehensive training set. This extensive dataset aimed to capture subtle variations, ensuring that the model could generalize well to different individuals and scenarios. By investing in thorough data collection and preprocessing, the system laid a strong foundation for achieving high accuracy and robustness in face recognition tasks.

Module 2.2: Siamese Neural Network Design and Training

The Siamese Neural Network architecture was implemented using TensorFlow to tackle similarity detection and one-shot learning challenges effectively. This architecture uses twin subnetworks with shared weights, ensuring both input images undergo identical transformations for extracting feature embeddings. By focusing on the similarity metric between embeddings, the network demonstrates robustness in tasks requiring high discriminative power.

Inspired by Koch et al., the network was trained using labeled image pairs with a triplet loss function. This objective optimizes the distance metric by minimizing intra-class distances while maximizing inter-class distances, ensuring reliable feature embedding separation. The dataset used consists of image pairs, structured similarly to benchmarks like Omniglot, which test the network's ability to generalize across unseen classes.



Initial training results indicated an accuracy of 92%, validating the network's potential. To further enhance performance, additional preprocessing techniques are being explored, including affine transformations, noise reduction, and augmentation strategies to simulate real-world variations. Advanced layers, such as attention mechanisms, are being integrated to refine feature extraction.

Building on Koch et al.'s insights, the project also plans to evaluate the network's transferability to tasks like signature verification and facial recognition. This expansion will help establish the broader applicability of the Siamese architecture in real-world scenarios requiring efficient similarity computation and robust one-shot learning capabilities.

Module 2.3: Attendance Monitoring System Integration

The Siamese Neural Network model has been successfully integrated into a prototype attendance monitoring system, marking a significant step forward in automating attendance tracking through facial recognition. The system is built using Python and leverages OpenCV for real-time face detection and TensorFlow for the comparison of face embeddings.

Currently, the system captures real-time images from a webcam, processes them through the trained Siamese network, and compares the detected faces with those stored in a database of registered faces. The database holds pre-registered face embeddings linked to user identities. The model then calculates the similarity score between the live image and the database images. If the similarity score exceeds a predefined threshold (e.g., 0.8), the system logs the attendance by recording the user's name and the timestamp of the match.

Key progress includes:

- Real-Time Face Detection: The system performs face detection and comparison at a rate of up to 15 frames per second, providing adequate performance for real-time monitoring in typical classroom or office settings.
- Attendance Logging: Once a match is found, the system automatically logs the user's attendance in the database, storing the user's identity along with the timestamp of the attendance.
- Database Integration: The system uses an SQLite database to store attendance records. Each entry includes the user's name and the exact time when their face was detected and matched. This process eliminates manual entry, providing an automated and efficient solution to track attendance.

The integration has been tested in controlled environments, and the system performs robustly under normal conditions, with further testing planned to address edge cases such as lighting variations and multiple faces within the frame. The system is currently undergoing optimization for speed and accuracy, with additional work aimed at improving frame processing rates and handling more complex real-world conditions.

Future improvements will focus on:

- Scalability: Enhancing the system to handle larger databases of registered faces and enabling multi-user recognition in larger spaces.
- Accuracy Tuning: Fine-tuning the model and optimizing its real-time performance to ensure reliable face matching even with variations in facial appearance (e.g., changes in lighting, angles, or facial expressions).
- UI Enhancements: Improving the user interface to allow users to easily view and export attendance records.

The attendance monitoring system is positioned to provide an efficient, accurate, and scalable solution to replace traditional manual attendance tracking systems, offering significant time savings and improved accuracy.

Chapter 3 Tasks Remaining

Phase 1: Attendance Matching Logic and Database Setup (Dec 20 - Jan 5)

1. **Task 1: Database Setup**
    - Set up a database to store user profiles and attendance logs.
    - Define the table structure for storing user information (e.g., name, face embeddings, attendance timestamps).
    - **Deadline**: **Dec 25**
2. **Task 2: Attendance Matching Logic**
    - Integrate the Siamese network for comparing real-time captured faces with stored face embeddings.
    - Implement a threshold (e.g., similarity score > 0.8) for marking attendance when a match is found.
    - **Deadline**: **Dec 30**
3. **Task 3: Storing Attendance in Database**
    - Develop functionality to store attendance logs with user names and timestamps once a match is confirmed.
    - Implement proper indexing and efficient query handling for retrieving attendance records.
    - **Deadline**: **Jan 5**

Phase 2: System Optimization and Testing (Jan 6 - Jan 20)

1. **Task 1: Performance Optimization**
    - Optimize the Siamese model for real-time inference.
    - Apply model pruning, quantization, or other techniques to improve speed without sacrificing accuracy.
    - **Deadline**: **Jan 10**
2. **Task 2: Test with Real Data**
    - Conduct live testing with real data, ensuring that the system can accurately capture faces and log attendance in different lighting and angle conditions.
    - **Deadline**: **Jan 15**
3. **Task 3: Edge Case Testing**
    - Test the system's performance in edge cases (e.g., multiple faces in the frame, low-light conditions, angle variations).

- Ensure system robustness across various real-world scenarios.
- **Deadline**: **Jan 20**

Phase 3: UI Enhancements and Integration Testing (Jan 21 - Feb 4)

1. **Task 1: UI Enhancements (Optional)**
   - If needed, improve the UI to show real-time status updates, like "Attendance Marked" or "Face Not Recognized" messages.
   - **Deadline**: **Jan 25**
2. **Task 2: Integration Testing**
   - Perform end-to-end testing for all integrated components:
     - Real-time image capture
     - Face comparison using the Siamese network
     - Database attendance logging
     - UI display of attendance records
   - **Deadline**: **Feb 4**

Phase 4: Final Testing, Documentation, and Deployment (Feb 5 - Mar 15)

1. **Task 1: Final System Testing**
   - Conduct thorough testing of the entire system under real-life conditions.
   - Test the system on multiple users to ensure correct attendance tracking and robust performance.
   - **Deadline**: **Feb 18**
2. **Task 2: Documentation**
   - Document the entire system, including setup instructions, system architecture, and deployment process.
   - Create user manuals for interaction with the UI and system maintenance.
   - **Deadline**: **Feb 25**
3. **Task 3: Deployment**
   - Deploy the attendance system to the production environment (local or cloud-based).
   - Perform final deployment checks and ensure the system runs smoothly.
   - **Deadline**: **Mar 15**

Development Plan: Attendance Monitoring System with Siamese Neural Network

Phase 1: Initial Setup and Model Integration (Nov 20 - Dec 3)

1. **Task 1: System Requirements Setup**
    - Install Python, TensorFlow, OpenCV, and other necessary libraries.
    - Set up the camera or image capture hardware for real-time input.
    - Deadline: Nov 24
2. **Task 2: Model Integration**
    - Load the pre-trained Siamese model and integrate it into the system.
    - Ensure the model accepts real-time images or video feed as input.
    - Deadline: Nov 28
3. **Task 3: Image Preprocessing**
    - Implement image preprocessing to resize and normalize incoming frames.
    - Ensure images are compatible with the Siamese model's input format.
    - Deadline: Dec 3

Phase 2: Attendance Matching Logic and Database Setup (Dec 4 - Dec 17)

1. **Task 1: Real-Time Face Capture**
    - Implement code for capturing real-time video frames using OpenCV.
    - Display the video feed for real-time monitoring.
    - Deadline: Dec 7
2. **Task 2: Database Setup**
    - Design and create an attendance database (SQLite or MySQL).
    - Define the table structure for storing user data and attendance logs.
    - Deadline: Dec 10
3. **Task 3: Attendance Matching**
    - Compare real-time captured faces with registered face embeddings.
    - Implement a matching threshold (e.g., similarity score of 0.8 or higher).
    - Deadline: Dec 17

Phase 3: System Logic and User Interface Development (Dec 18 - Jan 7)

1. **Task 1: Attendance Logging**
    - Develop the system to log attendance when a match is found.
    - Insert records into the attendance database with timestamps.
    - Deadline: Dec 22
2. **Task 2: Web Interface (Optional)**
    - Create a simple web interface using Flask or Django to display the attendance log.

- Display names and timestamps for when users’ attendance was marked.
- Deadline: Jan 7

Phase 4: System Optimization and Testing (Jan 8 - Jan 21)

1. **Task 1: Model Optimization**
    - Ensure the Siamese model performs optimally for real-time inference.
    - Adjust any hyperparameters or model components to improve speed and accuracy.
    - Deadline: Jan 14
2. **Task 2: Testing with Real Data**
    - Test the system with live input and attendance logging.
    - Ensure the attendance is accurately logged for different individuals.
    - Deadline: Jan 21

Phase 5: UI Enhancements and Integration (Jan 22 - Feb 4)

1. **Task 1: UI Improvements**
    - Improve the web interface by adding features like filtering, search, and pagination for attendance logs.
    - Deadline: Jan 28
2. **Task 2: Integration Testing**
    - Integrate all components: real-time image capture, face matching, database logging, and the user interface.
    - Perform end-to-end testing and resolve integration issues.
    - Deadline: Feb 4

Phase 6: Final Testing, Documentation, and Deployment (Feb 5 - Mar 15)

1. **Task 1: Final System Testing**
    - Conduct thorough testing for edge cases (e.g., multiple faces, low light conditions).
    - Test the system with different environments and users.
    - Deadline: Feb 18
2. Task 2: Documentation
    - Document the system architecture, code, and setup process.
    - Provide clear instructions for deployment and usage.
    - Deadline: Feb 28
3. Task 3: Deployment
    - Deploy the attendance system to a local server or production environment.
    - Conduct final deployment tests to ensure stability.
    - Deadline: Mar 15

Dates	Task
Nov 20 - Nov 24	System Requirements Setup
Nov 25 - Nov 28	Model Integration
Nov 29 - Dec 3	Image Preprocessing
Dec 4 - Dec 7	Real-Time Face Capture
Dec 8 - Dec 10	Database Setup
Dec 11 - Dec 17	Attendance Matching Logic
Dec 18 - Dec 22	Attendance Logging
Dec 23 - Jan 7	Web Interface Development
Jan 8 - Jan 14	Model Optimization
Jan 15 - Jan 21	Testing with Real Data
Jan 22 - Jan 28	UI Improvements
Jan 29 - Feb 4	Integration Testing
Feb 5 - Feb 18	Final System Testing
Feb 19 - Feb 28	Documentation
Mar 1 - Mar 15	Deployment

Fig 3. Current Development Plan

1.4 Conclusion

The project is currently ahead of schedule, with all required tasks and milestones being met within the planned timeframes. Significant progress has been made in developing the **Attendance Monitoring System** using **Siamese Neural Network Face Recognition**. Key components such as the **model training**, **real-time face detection**, and **database integration** have been successfully completed and integrated into the prototype.

As of now, the system is capable of accurately comparing registered face embeddings with live video input, logging attendance automatically. Future work will focus on **system optimization**, further **testing** (especially under different real-world conditions), and the final **deployment** of the application.

Given the current pace of development and the successful completion of major milestones, the project is expected to be finished on time. A fully functional demonstration of the **Attendance Monitoring System** will be available for the week of **April 15th, 2025**. This will include all final features, system optimizations, and comprehensive testing to ensure reliable performance.

The project is on track to deliver a robust and scalable solution for automated attendance tracking, with potential for further expansion and real-world deployment.

References

[1] Koch, G.R. (2015). Siamese Neural Networks for One-Shot Image Recognition.

[2] ***Jiankang Deng, Jia Guo, Evangelos Ververas, Irene Kotsia, Stefanos Zafeiriou***; Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 5203-5212

[3] Shifeng, Z., Xiangyu, Z., Zhen, L., Hailin, S., Xiaobo, W. & Stan, Z. L. (2018). FaceBoxes: A CPU Real-time Face Detector with High Accuracy. , .